

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: HOPE_DAC | **Context:** Global NLP/AI Research Community — Computational Mental Health (HOPE Benchmark Assessment)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Three expert linguists annotated the corpus [Q45], operating from a hierarchical scheme developed with mental health professional and linguist consultation [Q16, Q95], with calibration phase [Q46] and expert-therapist-moderated discussion [Q47]. Aggregate Cohen's Kappa is 0.7234 ('substantial') [Q48, Q49]. The authors transparently acknowledge that 'the annotator's bias cannot be ruled out completely' [Q96]. However, multiple critical gaps remain for the elicited HIGH-priority use case: (1) Annotator cultural backgrounds, native languages, and exposure to target downstream communities are entirely undocumented — the only reported demographics are age 25–35 and 2–10 years professional experience [Q50]; (2) No label-stratified IAA scores are reported, masking whether agreement varies on culturally loaded labels (GC, ACK, GT, ID); (3) The institutional affiliations (BITS Pilani, MAIT Delhi, IIIT-Delhi) [Q12] suggest annotators are likely South Asian by background — itself a confound when annotating US-clinical content for downstream targeting of US underrepresented communities; (4) Empirical evidence shows LLMs and transformer models (including RoBERTa, HOPE's backbone) exhibit racial and dialectal bias in clinical contexts [WEB-8, WEB-9], grounding the OC concern empirically. AnnoMI [WEB-15] demonstrates that richer annotator documentation is achievable in this field, making HOPE's gap a methodological choice rather than a constraint.

ID	Evidence
Q12	"1BITS Pilani, Goa, India, 2Maharaja Agrasen Institute of Technology, New Delhi, India, 3IIIT-Delhi, India." (p.1)
Q16	"To cater to the requirements of counseling conversations, we define a novel hierarchical annotation scheme for the dialogue-act annotation." (p.2)
Q45	"We employed three annotators who are experts in linguistics." (p.4)
Q46	"To ensure the understanding of the tasks and annotation scheme, we took a sample of the dataset and asked each annotator to annotate them as per the prepared set of guidelines." (p.4)
Q47	"Following this, every annotation was discussed in the presence of the annotators and an expert therapist as moderator to ensure consistency." (p.4)
Q48	"After the annotation process, we compute Cohen's Kappa score [7] to measure the agreement among annotators." (p.4)
Q49	"We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category." (p.4)
Q50	"Annotators are in the age group of 25-35, with 2-10 years of professional experience." (p.4)
Q95	"Moreover, we consulted with mental-health professionals and linguists in preparing the annotation guideline." (p.8)
Q96	"However, the annotator's bias cannot be ruled out completely." (p.8)
WEB-8	ResearchGate: RoBERTa shows complex bias toward AAVE in masked language modeling — bias inherited by SPARTA (https://www.researchgate.net/publication/371085866_Quantifying_the_Bias_of_Transformer-Based_Language_Models_for_African_American_English_in_Masked_Language_Modeling)
WEB-9	npj Digital Medicine (2025): LLMs more racially biased in mental health than other health domains; AAVE dialect cues trigger inferior treatment recommendations (https://www.nature.com/articles/s41746-025-01746-4)
WEB-15	AnnoMI uses 10 annotators with post-annotation survey, demonstrating richer annotator documentation is achievable (https://www.mdpi.com/1999-5903/15/3/110)

Strengths

- Substantial aggregate inter-annotator agreement ($\kappa=0.7234$) [Q49] indicates reasonable annotator consistency on the labels overall.
- Multi-stage annotation process with calibration [Q46] and expert-therapist-moderated discussion [Q47] follows good annotation methodology practice.
- Mental health professionals and linguists were consulted in guideline preparation [Q95].
- Authors transparently acknowledge that annotator bias cannot be ruled out [Q96] and that downstream effectiveness 'may vary' across regions [Q99].

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Q46	"To ensure the understanding of the tasks and annotation scheme, we took a sample of the dataset and asked each annotator to annotate them as per the prepared set of guidelines." (p.4)
Q47	"Following this, every annotation was discussed in the presence of the annotators and an expert therapist as moderator to ensure consistency." (p.4)
Q49	"We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category." (p.4)
Q95	"Moreover, we consulted with mental-health professionals and linguists in preparing the annotation guideline." (p.8)
Q96	"However, the annotator's bias cannot be ruled out completely." (p.8)
Q99	"Hence, the effectiveness of SPARTA on data from other geographical or demographical regions may vary." (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Substantially misaligned for target downstream communities. Ground truth labels reflect annotators applying a US-clinical-derived scheme [Q98] without documented exposure to or grounding in target communities' communicative norms.

OC-2: High potential for disagreement on culturally loaded utterances. Regional context documents that AAVE pragmatics, familismo opening, spiritual coping language, and indirect disclosure narratives are likely to be misclassified as GC, GT, OD, or ID by US-clinical-trained annotators. No empirical disagreement test exists [WEB-17].

OC-3: Annotator demographics are partially documented (age 25–35, 2–10 years experience [Q50]) but cultural background, native language, exposure to minority US clinical populations, and institutional clinical training are NOT DOCUMENTED — would need a Datasheet-style annotator demographics table.

OC-4: Re-annotation by a representative regional pool is appropriate for high-stakes downstream deployment. The elicitation (A2) frames this as complementary work. AnnoMI's 10-annotator design with post-annotation survey [WEB-15] demonstrates feasibility.

OC-5: Aggregation: only aggregate κ is reported [Q48, Q49]. No per-label IAA, no minority-perspective preservation mechanism documented. INSUFFICIENT DOCUMENTATION on whether disagreements were resolved by majority vote, expert moderator decision, or other mechanism beyond the qualitative 'group discussion' description [Q47].

OC-6: Documented label issues threatening convergent and external validity: (a) annotator cultural backgrounds undocumented; (b) no label-stratified IAA; (c) RoBERTa backbone [Q61] inherits AAVE bias [WEB-8]; (d) field-wide evidence that LLMs are more racially biased in mental health contexts than other domains [WEB-9, WEB-22]; (e) acknowledged residual

annotator bias [Q96].

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Q47	"Following this, every annotation was discussed in the presence of the annotators and an expert therapist as moderator to ensure consistency." (p.4)
Q48	"After the annotation process, we compute Cohen's Kappa score [7] to measure the agreement among annotators." (p.4)
Q49	"We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category." (p.4)
Q50	"Annotators are in the age group of 25-35, with 2-10 years of professional experience." (p.4)
Q61	"For each utterance Ut in a conversation dialogue, we extract the semantic representation through pre-trained RoBERTa model [25], which is subsequently utilized to leverage the local (Lt) and global (Gt) contextual information within the dialogue." (p.5)
Q96	"However, the annotator's bias cannot be ruled out completely." (p.8)
Q98	"Another important aspect of the current work is that the majority of the sessions in HOPE belong to the mental health professionals and patients based in United States." (p.8)
WEB-8	ResearchGate: RoBERTa shows complex bias toward AAVE in masked language modeling — bias inherited by SPARTA (https://www.researchgate.net/publication/371085866_Quantifying_the_Bias_of_Transformer-Based_Language_Models_for_African_American_English_in_Masked_Language_Modeling)
WEB-9	npj Digital Medicine (2025): LLMs more racially biased in mental health than other health domains; AAVE dialect cues trigger inferior treatment recommendations (https://www.nature.com/articles/s41746-025-01746-4)
WEB-15	AnnoMI uses 10 annotators with post-annotation survey, demonstrating richer annotator documentation is achievable (https://www.mdpi.com/1999-5903/15/3/110)
WEB-17	Translational Psychiatry (2023): only 4 of 102 NLP-MHI studies used external validation — subgroup performance reporting is field-wide rare (https://www.nature.com/articles/s41398-023-02592-2)
WEB-22	PMC (2024): documented racial bias in clinical LLMs (https://pmc.ncbi.nlm.nih.gov/articles/PMC12137607/)

Information Gaps

- Annotator cultural background, native language, race/ethnicity, and clinical exposure to target downstream communities.
- Label-stratified Cohen's Kappa scores.
- Disagreement resolution protocol beyond qualitative group discussion.
- Background of the moderating expert therapist.

Requires Expert Verification

- Re-annotation of a sample of HOPE utterances by annotators representative of South Asian, African American, and Latino communities to empirically test for systematic divergence in label assignment.

Remediation

Gap 1: Annotator cultural backgrounds and label-stratified IAA are not documented [Q50, Q49], and the RoBERTa backbone [Q61] inherits documented AAVE bias [WEB-8, WEB-9].

Recommendation: Run a re-annotation pilot on a sample of HOPE utterances using annotators representative of South Asian, African American, and Latino communities; compute label-stratified Cohen's Kappa and identify systematic divergences. Publish a Datasheet-style annotator demographics extension. Conduct an AAVE-specific bias audit on SPARTA following the protocols in [WEB-25].

Gap 2: Aggregation methods beyond qualitative 'group discussion' [Q47] for resolving annotator disagreement are not documented, risking erasure of minority annotator perspectives.

Recommendation: Document and report the disagreement resolution protocol explicitly; preserve and release individual annotator labels where feasible (following AnnoMI's [WEB-15] practice) to enable downstream researchers to model annotator disagreement as a signal rather than discarding it.

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Q47	“Following this, every annotation was discussed in the presence of the annotators and an expert therapist as moderator to ensure consistency.” (p.4)
Q49	“We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category.” (p.4)
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WEB-15	AnnoMI uses 10 annotators with post-annotation survey, demonstrating richer annotator documentation is achievable (https://www.mdpi.com/1999-5903/15/3/110)
WEB-25	https://aclanthology.org/2023.emnlp-main.421.pdf